

KE-05: failure of spectrum-uniform probabilistic interpolation bounds

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This manuscript gives a negative resolution of the original KE-05 probability bound. The conjecture and interpolation framework are due to Nian Shao. The proof uses the fixed canonical recurrence and deterministic real diagonal inputs.

Review and assistance. The [independent mathematical review](#) and [submission record](#) document the reviewed source and verification scope. The manuscript was developed with substantial AI assistance. An independent AI-agent audit is informal review; it does not establish external human peer review or formal verification. The coordinating agent contributed to the argument and is not counted as the independent reviewer.

Exact target and conclusion

The canonical KE-05 question fixes integers $b \geq 1$ and $d \geq 2$ and asks whether the random variable $\chi_{\text{mono}}\chi_{\text{coef}}$ defined by its stated recurrence is uniformly bounded in probability over all real diagonal blocks with pairwise disjoint spectra. In particular, for each $0 < \delta < 1$ there should be a finite $C(b, d, \delta)$ that works for every fixed admissible diagonal input. All matrix norms are spectral norms, and all entries of the square matrices Ω_i are independent standard real Gaussian variables.

The answer to this exact question is **no**. The failure occurs already for $b = 2$ and $d = 3$, with deterministic admissible inputs

$$\Lambda_1 = 2I_2, \quad \Lambda_2(\varepsilon) = \begin{pmatrix} \varepsilon & 0 \\ 0 & 2\varepsilon \end{pmatrix}, \quad \Lambda_3 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad 0 < \varepsilon < \frac{1}{4}.$$

Their spectra are pairwise disjoint. The repeated eigenvalue within the first block is permitted by the canonical hypotheses. For every finite real number C ,

$$\lim_{m \rightarrow \infty} \Pr\{\chi_{\text{mono}}(\varepsilon_m)\chi_{\text{coef}}(\varepsilon_m) \leq C\} = 0, \quad \varepsilon_m = \frac{1}{m+5}, \quad m = 1, 2, \dots$$

Thus even the choice $\delta = 1/2$ admits no finite uniform constant. The proof uses the literal canonical recurrence, without relying on a matrix-polynomial factorization theorem.

1. Recurrence and the particular ordering

For completeness, in any of the orderings prescribed in KE-05, the recurrence is computed in descending order of i :

$$B_i = \Omega_i^{-1} \Lambda_i \Omega_i, \quad S_{i,i} = I_2, \quad S_{i,j} = B_i S_{i,j-1} - S_{i,j-1} \widehat{B}_j \quad (j = i+1, \dots, 3),$$

$$\widehat{\Omega}_i = \Omega_i S_{i,3}, \quad \widehat{B}_i = \widehat{\Omega}_i^{-1} \Lambda_i \widehat{\Omega}_i = S_{i,3}^{-1} B_i S_{i,3}.$$

The superscript (k) denotes the ordering whose first original block is k , as in the canonical statement. The global quantities are the maxima of their respective ordering-specific quantities.

We obtain lower bounds from ordering $k = 1$, which is the original ordering $(1, 2, 3)$, and omit its superscript in intermediate formulas. The global endpoints for our diagonal input are $a = 0$ and $c = 2$. Consequently, the $i = 2$ term in the definition of the first maximum gives

$$\chi_{\text{mono}} \geq \chi_{\text{mono}}^{(1)} \geq \frac{\|\widehat{B}_2\|_2}{\|\Lambda_2(\varepsilon)\|_2} = \frac{\|\widehat{B}_2\|_2}{2\varepsilon}.$$

No zero-over-zero convention is involved in this particular ratio.

The first-block minimum cross-gap is exactly one:

$$\min_{\substack{i=2,3, \\ \lambda \in \sigma(\Lambda_1) \\ \eta \in \sigma(\Lambda_i)}} |\lambda - \eta| = \min\{2 - \varepsilon, 2 - 2\varepsilon, 2, 1\} = 1.$$

Since $d - 1 = 2$, it follows that

$$\chi_{\text{coef}} \geq \chi_{\text{coef}}^{(1)} = \|S_{1,3}^{-1}\|_2^{1/2}.$$

2. An almost-sure nonzero limit of the transformed second block

Work first on the event that Ω_2 and Ω_3 are invertible, which has probability one. Put

$$X = \Omega_2^{-1} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \Omega_2, \quad P = \Omega_3^{-1} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \Omega_3.$$

Then $B_2 = \varepsilon X$, $B_3 = P$, $\det X = 2$, and

$$P^2 = P, \quad \text{tr } P = 1, \quad \det P = 0, \quad \text{adj } P = I_2 - P.$$

Here adj is the classical adjugate. Define the real random scalar and matrix

$$\kappa = \text{tr}((I_2 - P)X), \quad Q = (I_2 - P)XP.$$

The last block is unchanged by the recurrence, so

$$\widehat{B}_3 = P, \quad S_{2,3} = \varepsilon X - P, \quad \widehat{B}_2 = (\varepsilon X - P)^{-1} \varepsilon X (\varepsilon X - P).$$

The elementary 2×2 determinant expansion yields

$$\det(\varepsilon X - P) = \varepsilon^2 \det X - \varepsilon \text{tr}((\text{adj } P)X) + \det P = \varepsilon(2\varepsilon - \kappa).$$

The adjugate is linear on 2×2 matrices. Hence, whenever this determinant is nonzero, cancellation of the scalar factor ε gives the exact formula

$$\widehat{B}_2 = \frac{(\varepsilon \text{adj } X - (I_2 - P))X(\varepsilon X - P)}{2\varepsilon - \kappa}.$$

On the event $\kappa \neq 0$, this formula is defined for all sufficiently small positive ε and has the limit

$$\widehat{B}_2 \rightarrow N := -\frac{(I_2 - P)XP}{\kappa} = -\frac{Q}{\kappa} \quad (\varepsilon \downarrow 0).$$

The identity $P(I_2 - P) = 0$ also gives $N^2 = 0$. We next show that this limiting nilpotent matrix is nonzero almost surely; it is not being assumed to be nonzero merely because the recurrence is nonsingular.

The entries of X and P , and therefore κ and the entries of Q , are rational functions of the eight real entries of Ω_2, Ω_3 . Their denominators can be cleared by products of determinants of these two matrices. Consider the invertible rational matrices

$$\Omega_2 = I_2, \quad \Omega_3 = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}.$$

At this single deterministic witness,

$$X = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad P = \begin{pmatrix} 6 & 10 \\ -3 & -5 \end{pmatrix}, \quad \kappa = 7, \quad Q = \begin{pmatrix} 30 & 50 \\ -18 & -30 \end{pmatrix}.$$

In particular, the numerator polynomials of κ and of the $(1, 1)$ entry of Q are not identically zero. A nonzero polynomial in finitely many real variables has a Lebesgue-null zero set. One elementary proof is induction on the number of variables: outside the common zero set of the coefficient polynomials, each univariate slice has finitely many roots, and Fubini's

theorem finishes the induction. Since the joint Gaussian law has a density, each of these zero sets has probability zero. Together with almost-sure invertibility of the two Gaussian matrices, this proves

$$\Pr\{\kappa \neq 0, Q_{11} \neq 0\} = 1.$$

It follows that

$$\|\widehat{B}_2(\varepsilon)\|_2 \rightarrow \|N\|_2 > 0 \quad \text{almost surely as } \varepsilon \downarrow 0$$

along values where the recurrence is defined. The matrix-norm limit uses only continuity of the spectral norm.

For an additional exact check at the displayed rational witness,

$$\widehat{B}_2 = \frac{1}{2\varepsilon - 7} \begin{pmatrix} 2\varepsilon^2 - 7\varepsilon + 30 & 20\varepsilon + 50 \\ 3\varepsilon - 18 & 4\varepsilon^2 - 14\varepsilon - 30 \end{pmatrix},$$

and

$$N = \frac{1}{7} \begin{pmatrix} -30 & -50 \\ 18 & 30 \end{pmatrix}, \quad \|N\|_2 = \frac{68}{7}.$$

The witness is used to prove that the relevant polynomials are nonzero. The counterexample does not require drawing that particular Gaussian outcome, an event of probability zero.

3. A deterministic lower bound for the coefficient factor

Because $\Lambda_1 = 2I_2$, we have $B_1 = 2I_2$ regardless of Ω_1 . The literal recurrence therefore gives

$$S_{1,2} = 2I_2 - \widehat{B}_2, \quad S_{1,3} = 2S_{1,2} - S_{1,2}P = (2I_2 - \widehat{B}_2)(2I_2 - P).$$

This particular factorization uses the scalar identity $B_1 = 2I_2$ and does not reorder two noncommuting factors. Whenever $S_{2,3}$ is invertible, $\widehat{B}_2$ is similar to $\Lambda_2(\varepsilon)$. Thus

$$\det S_{1,3} = \det(2I_2 - \widehat{B}_2) \det(2I_2 - P) = 2(2 - \varepsilon)(2 - 2\varepsilon).$$

For $0 < \varepsilon < 1/4$, this determinant is positive and less than eight. In particular, $S_{1,3}$ is invertible without any further restriction on the Gaussian outcome for this ordering.

For any invertible 2×2 matrix S , its singular values $s_1 \geq s_2 > 0$ satisfy $s_2^2 \leq s_1 s_2 = |\det S|$. Consequently

$$\|S^{-1}\|_2 = \frac{1}{s_2} \geq |\det S|^{-1/2}.$$

Using the unit cross-gap established in Section 1, we obtain

$$\chi_{\text{coef}} \geq \|S_{1,3}^{-1}\|_2^{1/2} \geq [2(2 - \varepsilon)(2 - 2\varepsilon)]^{-1/4} \geq 8^{-1/4}.$$

In particular, this factor cannot cancel the divergence of the monomial factor.

4. All prescribed orderings are defined almost surely

We give the short genericity argument explicitly so that this proof does not need a nonsingularity theorem about matrix-polynomial factorizations.

Fix any one value $0 < \varepsilon < 1/4$ and any ordering required by KE-05. The entries produced by the finite recurrence are rational functions of the entries of the three matrices Ω_i as long as the preceding inverses exist. Set $\Omega_1 = \Omega_2 = \Omega_3 = I_2$. For this deterministic witness all B_i are diagonal, and induction down the recurrence gives

$$\widehat{B}_i = B_i = \Lambda_i, \quad S_{i,j} = \prod_{\ell=i+1}^j (\Lambda_i - \Lambda_\ell)$$

in the chosen ordering. The empty product is I_2 . Every diagonal entry of each factor is nonzero, since the spectra of different blocks are disjoint. Thus every matrix whose inverse is required is invertible at this witness.

Proceeding through the finite list of inverses, each determinant, written as a rational function with the previously encountered denominators cleared, has a numerator polynomial that is nonzero at this witness. The exceptional set is therefore contained in a finite union of zero sets of nonzero polynomials. It is Lebesgue-null and hence has Gaussian probability zero. This applies to the finitely many root orderings simultaneously.

Apply this argument to each member of the deterministic sequence $\varepsilon_m = 1/(m+5)$, $m \geq 1$. A countable union of null sets is null, so with probability one every prescribed ordering is defined for every member of this sequence. The endpoint convention in the canonical question makes any scalar-block 0/0 ratio equal to one; it does not affect the lower bounds above.

5. Failure of every finite uniform probability bound

Couple the inputs at different ε_m by using the same three Gaussian matrices $\Omega_1, \Omega_2, \Omega_3$ at every index. For each fixed index this coupling has exactly the distribution specified in KE-05: the diagonal input is deterministic and the entries of all three matrices are independent standard Gaussians.

On a probability-one event, the conclusions of Sections 2 and 4 hold together. On that event the global constants are defined for every m and satisfy

$$\chi_{\text{mono}}(\varepsilon_m) \chi_{\text{coef}}(\varepsilon_m) \geq 8^{-1/4} \frac{\|\widehat{B}_2(\varepsilon_m)\|_2}{2\varepsilon_m} \rightarrow +\infty.$$

For any finite real C , the indicators of the events that these products are at most C therefore converge almost surely to zero. They are bounded by one, so the dominated convergence theorem yields

$$\Pr\{\chi_{\text{mono}}(\varepsilon_m) \chi_{\text{coef}}(\varepsilon_m) \leq C\} \rightarrow 0.$$

Changing the constants arbitrarily on the probability-zero sets where an inverse is undefined does not change this conclusion.

If the asserted $C(2, 3, 1/2)$ existed, each probability in the last display with that finite threshold would be at least $1/2$, a contradiction. This disproves the canonical uniform-in-eigenvalues claim.

The coupling is only a proof device for the marginal-probability limit. No input eigenvalue depends on a sampled Gaussian matrix, and no stronger requirement that one Gaussian draw succeed for every spectrum is imposed on the conjecture.

Scope, attribution, and verification limits

The negative result concerns the random constants defined by the literal canonical KE-05 recurrence. Its input uses interlaced block spectra: the eigenvalues of Λ_2 lie between those of Λ_3 . Pairwise disjoint spectra permit this. The argument makes no assertion about a different question imposing ordered, disjoint spectral intervals on the blocks, and it does not disprove cluster-robust convergence of a block Krylov algorithm itself.

Nian Shao introduced the source interpolation framework and the conjectured spectrum-independent probabilistic bounds in *A structural bound for cluster robustness of randomized small-block Lanczos*, arXiv:2507.10144v2, 30 May 2026; relevant locators are the recurrence in Lemma 3, equation (12), and Conjecture 1 in Section 3.2. The canonical problem and these source locations were read for scope. The proof here consists of elementary 2×2 identities, Gaussian absolute continuity, and dominated convergence. It does not depend on the correctness of a general ordered-product identity in that paper.

The accompanying exact-arithmetic checker verifies the displayed rational witness, polynomial identities at that witness, and selected finite-input recurrence instances. Those checks support transcription and algebra; the general probability proof is the argument above. The linked independent review and public eligibility audit document the separate mathematical and publication checks.

Primary references

1. Canonical KE-05, *Spectral-gap-independent constants for randomized block polynomial interpolation*: <https://github.com/ajt60gaibb/OpenProblemsInNLA/blob/main/randomized-and-low-rank-approximation/KE-05/README.md>.
2. N. Shao, *A structural bound for cluster robustness of randomized small-block Lanczos*, arXiv:2507.10144v2, 30 May 2026: <https://arxiv.org/html/2507.10144v2>.